

Mind-Body TaoRelax: Relieving Stress through Immersive Virtual Reality Relaxation Training in a Taoist Atmosphere

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ABSTRACT

Virtual reality technology is providing a new method for psychological relaxation training, but most systems focus primarily on meditation as a singular relaxation method. We propose a novel virtual reality relaxation system called TaoRelax. Building upon traditional Taoist culture, the system integrates both meditation and progressive muscle relaxation training. It combines electromyography (EMG) as a means of feedback and assessment during training, providing users with a dual experience of relaxation for both body and mind. Our preliminary results indicate that after training in our system, participants experienced a significant reduction in psychological stress and a notable improvement in meditation abilities.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

With the continuous advancement of virtual reality (VR) and augmented reality (AR) technologies, the opportunity to integrate psychological relaxation with virtual environments for training is gradually emerging [1, 2, 6, 10]. In the VR world, a high level of immersion allows us to escape disturbances such as environmental noise in real life. Furthermore, compared to simple audio or human guidance on a screen, VR can provide rich scenarios and visual effects that are challenging to replicate in the real world, thus enhancing the effectiveness of psychological relaxation training.

In Chinese Taoist culture, meditation is regarded as a spiritual cultivation method to achieve inner tranquility and spiritual growth through concentration, contemplation, and self-reflection [7, 13]. Taoism emphasizes the harmony between individuals and the universe, the natural balance, and inner cultivation. For beginners, immersing themselves in Taoist elements can help them enter a state of mental relaxation more effectively [3]. Unlike meditation focused on relaxing the mind, progressive muscle relaxation training starts with relaxing muscles and subsequently promotes mental relaxation, addressing mental stress-induced muscle tension [5, 8, 11].

Considering these, we develop an immersive virtual reality psychological relaxation system to assist users in learning mindfulness meditation and progressive muscle relaxation training. The system integrates elements from Chinese Taoist Tai Chi, among others, aiding users in a deep understanding of the connection between the body and mind, as well as the relationship between humans and nature. Additionally, the system combines the changes in two physiological data, EMG and heart rate, with variations in scene elements, providing users with real-time biofeedback functionality.

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2 SYSTEM

The system we developed is a virtual reality-based psychological relaxation platform using the Unity3D engine. In order to provide users with enhanced relaxation effects, the system offers instruction and guidance on two types of mental relaxation techniques.

2.1 System Scenes

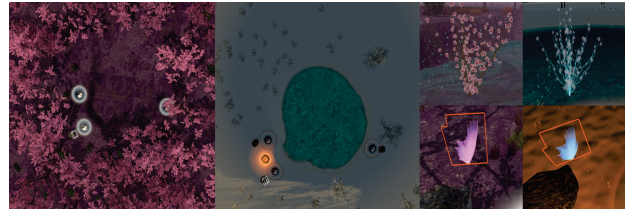


Figure 1: System Scenes.

In order to immerse users in a relaxed environment within the virtual world, the system introduces two distinct virtual natural environments: Cherry Blossom Forest and Snowy Mountain Plains (see Fig. 1). To enrich the user experience, both relaxation scenes simulate different times of the day. The Cherry Blossom Forest scene simulates noon, while the Snowy Mountain Plains scene simulates early morning.

The elements of both scenes are deeply infused with Chinese Taoist culture. In front of each user's virtual avatar stands a corresponding Tai Chi disk. The rotation and scaling of this disk guide each inhalation and exhalation, considering breath as a bridge for the exchange of yin and yang energy. In the center of the lakes in both scenes, vines steadily grow. As the training progresses, these vines continuously expand, causing the water level in the lakes to rise, reflecting vitality and the harmonious symbiosis between humans and nature. As users delve into meditation, letting go of all distractions and perceiving themselves and the world from the third-person perspective of a butterfly, these effects are illustrated in Fig. 1.

2.2 Physiological Data Feedback

The system incorporates various forms of feedback based on EMG and heart rate into the virtual scenes to assist users in relaxation training.

EMG Feedback EMG is a physiological data form characterized by significant randomness and fluctuations. To enhance users' perception of their own EMG changes, the system integrates real-time EMG data with fountains in the virtual environment (see Fig. 2). The height of the fountain corresponds to the real-time size of the user's EMG data.

Heart Rate Feedback Within the system, each user's virtual avatar is equipped with a corresponding Tai Chi disk (see Fig. 3a). The rotation and scaling of the Tai Chi disk guide the user's breathing. Additionally, the outer edge of the Tai Chi disk has three luminescent



Figure 2: Particle fountain.

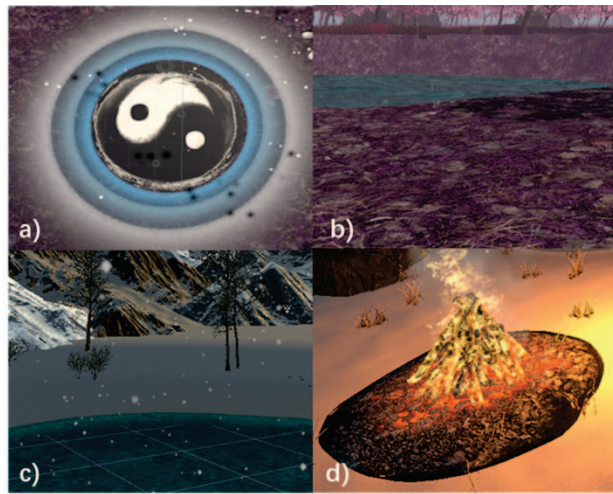


Figure 3: Tai Chi disk, raindrops, snowflakes and the flicker of a bonfire.

rings transitioning from blue to white. The user's heart rate is divided into three layers, each lower layer represented by an additional luminous ring. In addition to this visually intuitive halo effect, users can perceive the rhythmic pulsation of a cartoon heart, synchronized with their heart rate. Furthermore, raindrops in the Cherry Blossom Forest (see Fig. 3b), snowflakes in the Snowy Mountain Plains (see Fig. 3c), and the flickering of a bonfire (see Fig. 3d) dynamically respond to changes in the user's heart rate.

3 USER STUDY

This study involved a total of 25 participants, all of whom were recruited throughout the university campus via e-mail or approached personally, including 19 males and 6 females. The age range of the participants was 20 to 24 years old ($M = 21.68$, $SD = 0.80$). Among the participants, 72% had no prior experience with VR devices. The pre-experimental anxiety scores of all participants are within normal levels. The study was approved by the university's ethics committee (ID:2021ZDKYSB207).

The user study comprised two consecutive experimental phases. Participants initially underwent the first relaxation session in the Cherry Blossom Forest scene, followed by the second relaxation session in the Snowy Mountain Plains scene after a period of two to three days. We utilized the OYMotion dry electrode EMG sensor to capture surface EMG signals from muscles [12], and heart rate was monitored using the Garmin HRM Pro heart rate monitor [9].

In this study, we designed an anxiety induction phase, requiring all

participants to watch a VR video before the psychological relaxation training in the first experiment. This video was sourced from anxiety-inducing videos in OpenLAV [4], and it also included a self-authored half-minute video clip depicting anxious university students.

4 RESULTS

Our preliminary results indicate that after training in our system, participants experienced a significant reduction in psychological stress and a notable improvement in meditation abilities. In summary, this study demonstrates the effectiveness and usability of our TaoRelax system. The system incorporates physiological data feedback and integrates scenes infused with Taoist culture. It offers immersive VR scenarios and assists participants in achieving relaxation of both the mind and body through a combination of meditation and Progressive Muscle Relaxation. Additionally, the study explores the auxiliary effects of EMG in psychological relaxation. In essence, this study provides valuable insights into practical avenues for virtual reality-assisted psychological relaxation methods, simultaneously paving the way for the integration of culture and technology in the realm of mental health.

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